Personal Expense Tracker with Charts — Full Stack Development Mini Project

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1. Title Page

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================================================================================
A MINI-PROJECT REPORT ON PERSONAL EXPENSE TRACKER WITH CHARTS (MONETA)
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Submitted in partial fulfillment of the requirements for the award of the Degree of

BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE AND ENGINEERING (Full Stack Development Stream)

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2. Student Details

Attribute	Details
Candidate Name	[Student Full Name Placeholder]
University Roll No. / USN	[University Roll Number / USN Placeholder]
Semester & Section	[Semester VII / Section A Placeholder]
Department	Department of Computer Science & Engineering
Institution / College	[College / University Name Placeholder]
Affiliated University	[Affiliating University Name Placeholder]
Academic Session	2025 – 2026
Faculty Guide / Supervisor	[Prof. Guide Name Placeholder, Designation, Dept]

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3. Certificate

```
================================================================================
CERTIFICATE
================================================================================
```

This is to certify that the project entitled "PERSONAL EXPENSE TRACKER WITH CHARTS (MONETA)" is a bonafide work carried out by [Student Name Placeholder] (USN: [Roll Number Placeholder]) in partial fulfillment of the academic curriculum for Full Stack Development Mini-Project during the academic year 2025-2026.

The project has been reviewed, evaluated, and approved for submission.

_____ _____ Faculty Project Guide Head of Department (CSE) [Designation & Department] Department of Computer Science Date: _____ Date: _____

_____ _____ Internal Examiner External Examiner Date: _____ Date: _____

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4. Declaration

I hereby declare that this mini-project report entitled **"Personal Expense Tracker with Charts (MONETA)"** submitted to the Department of Computer Science and Engineering is a record of original work done by me under the supervision of my faculty guide.

The application logic, architecture diagrams, tests, and documentation are authentic and have not previously formed the basis for the award of any degree, diploma, or fellowship at this or any other university.

Date: October 25, 2026 **Place:** [City / Institution Location Placeholder]

(Signature of Candidate) **Name:** [Student Full Name Placeholder] **USN / Roll No:** [University Roll Number Placeholder]

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5. Acknowledgment

I express my deepest gratitude to my Project Guide, [Guide Name Placeholder], for their invaluable encouragement, technical guidance, and constructive critiques throughout the design, implementation, and verification phases of this application.

I extend my appreciation to the Head of the Department, [HOD Name Placeholder], and all faculty members of the Department of Computer Science & Engineering for providing the laboratory facilities and academic ecosystem necessary to complete this project.

Finally, I thank my family and peers for their continuous support and collaboration during the development of this Full Stack Development mini-project.

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6. Abstract

Personal financial awareness is essential for individuals navigating high-velocity digital payment ecosystems such as the Unified Payments Interface (UPI) and digital card networks. However, most consumers struggle with cumbersome manual spreadsheets or third-party cloud apps that raise severe data privacy risks.

This project presents **MONETA**, a client-side personal finance tracking and analytical web application designed with an editorial aesthetic inspired by Linear and Stripe. Engineered using **React 18, TypeScript 5.6, Tailwind CSS 3.4, and Vite 5**, MONETA delivers reactive financial intelligence without external cloud latency or data transmission. The platform incorporates a strict mathematical engine that computes opening balances, period net savings, historical savings velocity (%), and deterministic anomaly insights directly from client data. Dynamic visualizations powered by **Recharts** feature dual-gradient Area charts, category donut charts with centered balance counters, and longitudinal line plots.

Data is securely managed via a typed, fault-tolerant **Web Storage (LocalStorage) engine** supporting JSON backup restoration and CSV export. The system includes an automated test suite implemented in **Vitest and React Testing Library** verifying calculation formulas, persistence layers, and UI components across 15 passing test suites. MONETA establishes that an autonomous client-side application can provide institutional-grade design, mathematical rigor, and complete data privacy for modern personal finance management.

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8. Introduction

Managing discretionary and fixed financial outlays is a fundamental requirement of personal economic well-being. With the proliferation of instant mobile payments in India (UPI volume exceeding billions of transactions monthly), individuals execute dozens of micro-transactions daily. As a result, maintaining a clear mental ledger of cumulative expenditures, category limits, and net savings rates has become increasingly difficult.

The **Personal Expense Tracker with Charts (MONETA)** was developed as an undergraduate Full Stack Development (FSD) mini-project to resolve this challenge. Built using a reactive component-driven architecture, MONETA acts as an autonomous personal finance command center. It eliminates the traditional requirement for costly cloud backend servers by leveraging modern browser APIs, strict TypeScript domain contracts, and deterministic analytical heuristics.

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9. Problem Statement

Personal financial record-keeping currently suffers from three major structural flaws: 1. **Spreadsheet Inefficiency:** Traditional spreadsheet workbooks (Excel, Google Sheets) require manual formula maintenance, do not adapt comfortably to mobile touchscreens, and lack interactive visualizations for daily expenditure tracking. 2. **Telemetry and Privacy Invasion:** Commercial personal finance SaaS platforms demand access to SMS inboxes, email parsers, or banking credentials, transmitting private spending patterns to third-party ad networks. 3. **Data Disconnection in Basic Trackers:** Most educational mini-projects display static dummy cards or hardcoded chart arrays disconnected from real transaction registers. This leads to mathematical inconsistencies between displayed totals and underlying records.

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10. Objectives

- **Unified Architectural Consistency:** Build a unified single-page application (SPA) where adding, modifying, or removing a transaction immediately updates all dashboard counters, category donut charts, cash-flow gradients, and budget limits. - **Mathematical Integrity:** Enforce exact accounting standards: $$\text{Net Balance} = \text{Opening Balance} + \sum \text{Inflow} - \sum \text{Outflow}$$ $$\text{Savings Rate} = \frac{\sum \text{Inflow} - \sum \text{Outflow}}{\sum \text{Inflow}} \times 100$$ - **Editorial UI/UX Standard:** Implement a design system with warm neutral backgrounds (#F8F9FA), pure white cards, dark-mode switching, tabular numeral

alignment, and responsive viewports (1440px to 320px). - **Zero-Latency Client Persistence:** Provide local persistence with schema validation, CSV report generation, and JSON backup restoration. - **Automated Verification:** Achieve 100% automated pass rates across all financial formulas and storage modules using Vitest.

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11. Scope of the Project

In Scope

- Client-side transaction management (Income and Expenses across 18 standard Indian fintech categories). - Period filtering (This Month , Last Month , Last 3 Months , This Year , All Time). - Multi-dimensional transaction filtering by nature, category, payment channel, and search keywords. - Dynamic Recharts data visualizations (Area, Donut, Bar, and Line charts). - Category budget guardrails with visual threshold states and daily safe-to-spend allowance calculations. - Display currency customization across 9 global currencies with Indian Rupee (INR ₹) as primary default. - Data export (CSV, JSON) and backup restoration. - Automated testing suite and CI-ready test scripts.

Out of Scope (Documented Boundary)

- Live server-side bank account scraping (preventing regulatory and API credential overhead). - Real-time live foreign exchange (FX) market trading feeds.

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12. Existing System and Limitations

| Parameter | Traditional Manual Spreadsheets | Commercial Cloud Budget Apps | Existing Student Mini-Projects | | :--- | :--- | :--- | :--- | | **Setup Friction** | High (manual formula authoring) | High (mandatory phone/account sign-up) | Low | | **User Privacy** | High (local files) | Low (data monetized / synced to cloud) | Moderate | | **Visual Charts** | Basic, static Excel charts | Rich, but cluttered with upsells/ads | Hardcoded, disconnected mockups | | **Mobile UX** | Poor cell tap targets | Proprietary app install required | Unresponsive / horizontally clipped | | **Calculations** | Manual formulas (vulnerable to error) | Cloud-computed black box | Fake random numbers |

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13. Proposed System

MONETA provides an autonomous, zero-configuration web application running entirely inside the user's browser runtime. It guarantees: 1. **Total Data Sovereignty:** All transaction ledgers and budget targets reside strictly within the client's browser LocalStorage. No telemetry leaves the machine. 2. **Instant Reactivity:** State transitions occur in sub-millisecond execution times utilizing React's virtual DOM reconciliation and React Context dispatchers. 3. **Deterministic Heuristics:** Financial insights (such as primary category drain, month-over-month percentage changes, and budget depletion warnings) are derived from pure mathematical rules rather than probabilistic AI models. 4. **Resilient Data Portability:** Users can backup their full ledger in standard JSON and export records in comma-separated CSV format.

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14. Functional Requirements

- **FR-1: Transaction Ingestion:** The system shall validate and capture transaction description, monetary amount (>0), transaction type (income/expense), category, date (YYYY-MM-DD), payment method, and optional notes. - **FR-2: Ledger Audit & Pagination:** The ledger shall present transactions with sorting (date and amount) and paginate records at 10 items per page with real-time summary indicators. - **FR-3: Multi-Criteria Filtering:** The application shall permit real-time filtering across text search, transaction nature, category classification, and payment instruments without network requests. - **FR-4: Financial Aggregation:** The system shall continuously calculate total income, total outlays, net savings, capital balance, and savings rate across designated temporal intervals. - **FR-5: Budget Enforcement:** Users shall be able to define monthly spending ceilings globally and per category. The system shall reject duplicate active budgets for identical category periods. - **FR-6: Threshold Alerts:** The application shall evaluate and categorize budget utilization into three visual states: Safe ($<80\%$), Warning ($80\% - 99\%$), and Exceeded ($\ge 100\%$). - **FR-7: Safe Daily Spend Allowance:** The system shall compute daily allowable expenditure for the remainder of the active calendar month: $$\text{Daily Allowance} = \frac{\max(0, \text{Limit} - \text{Spent})}{\max(1, \text{Days in Month} - \text{Current Day} + 1)}$$ - **FR-8: Data Portability:** The application shall support JSON file import, JSON complete state export, and CSV table generation. - **FR-9: Theme & Currency Selection:** Users shall be capable of toggling Light/Dark/System themes and selecting display currency notations.

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15. Non-Functional Requirements

- **NFR-1: Performance & Velocity:** UI interactions, chart re-renders, and filtering updates shall execute in $<100\text{ ms}$. - **NFR-2: Usability & Design:** Visual layouts

shall comply with modern typography standards (Inter sans-serif, tabular figures, high contrast WCAG 2.1 AA compliance). - **NFR-3: Reliability:** The storage engine shall catch malformed records and provide fallback seed data rather than crashing the interface. - **NFR-4: Responsiveness:** Layouts shall reflow gracefully from 1440px desktop screens down to 320px narrow mobile screens without horizontal scroll. - **NFR-5: Maintainability:** Modules shall follow clear separations: domain types (/types), financial math (/utils), UI components (/components), and pages (/pages).

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16. Hardware and Software Requirements

Development & Host Environment

- **Processor:** Intel Core i5 / Apple Silicon M1 or higher - **RAM:** 8 GB minimum - **Disk Storage:** 500 MB free space - **Operating System:** macOS Sonoma / Linux Ubuntu 22.04 / Windows 11 - **Node.js:** v18.0.0+ (Tested on Node v26.8.1) - **Package Manager:** npm v9+

Client Browser Runtime

- Modern HTML5/ES2020-compliant browser: Google Chrome 90+, Mozilla Firefox 88+, Apple Safari 14+, Microsoft Edge 90+.

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17. Technology Stack and Rationale

```
+-------------------------------------------------------------+
| PRESENTATION LAYER |
| React 18.3 TypeScript 5.6 Tailwind CSS 3.4 |
+-------------------------------+-----------------------------+
| DATA VISUALIZATION |
| Recharts 2.13 (SVG Area, Pie, Bar) |
+-------------------------------+-----------------------------+
| STATE MACHINE & APPLICATION LOGIC |
| AppContext ToastContext React Hook Form |
+-------------------------------+-----------------------------+
| PERSISTENCE & CLIENT DATABASE |
| Browser Web Storage API (LocalStorage) |
+-------------------------------+-----------------------------+
```

- **React 18 & Vite:** Chosen for declarative component-driven UI and lightning-fast developer experience with zero bundling overhead during development. - **TypeScript:** Guarantees compile-time safety across transactions, budgets, settings, and utility calculation contracts. - **Tailwind CSS:** Provides atomic, maintainable design styling without CSS stylesheet bloat. - **Recharts:** Built on React component wrappers over D3

SVG elements, enabling responsive and accessible chart rendering. - **Zod & React Hook Form:** Provides type-safe form validation with minimal re-renders.

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18. System Architecture and Data Flow

```mermaid
flowchart TD
    User([User Action / Input]) --> UI[React UI Components]
    UI --> Forms[React Hook Form + Zod Schema]
    Forms --> AppCtx[AppContext State Manager]
    AppCtx --> Storage[LocalStorage Data Access Layer]
    Storage --> Seed[Initial Realistic Indian Seed Data]
    AppCtx --> Calcs[Financial Calculation Engine]
    Calcs --> Sum[Financial Summary KPI Cards]
    Calcs --> ChartArea[Cash Flow Velocity Area Chart]
    Calcs --> ChartDonut[Category Spending Donut Chart]
    Calcs --> Budgets[Budget Progress & Safe Daily Cap]
    Calcs --> Insights[Deterministic Rule Insights]
    Storage -.-> BackupJSON[(JSON Backup File)]
    Storage -.-> ExportCSV[(CSV Spreadsheet File)]
```

Data Flow Pipeline

1. The user inputs a transaction or alters a period filter. 2. The form validates constraints through the Zod schema. 3. The mutation is committed to the central AppContext state and persisted synchronously to LocalStorage. 4. Dependent reactive functions in calculations.ts recompute income sums, expense sums, savings rates, and budget statuses. 5. React triggers an optimized render update to the UI cards and SVG charts within a single event loop tick.

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19. Module Descriptions

1. **src/types/index.ts**: Contains domain entity interfaces (Transaction, Budget, UserSettings, FinancialSummary, BudgetStatus, SpendingInsight). 2. **src/utils/calculations.ts**: Pure mathematical utility library containing: - calculateTotalIncome() - calculateTotalExpenses() - calculateSavingsRate() - calculatePercentageChange() - aggregateExpensesByCategory() - aggregateTransactionsTimeline() - calculateBudgetStatuses() - generateSpendingInsights() 3. **src/utils/storage.ts**: Encapsulates Web Storage serialization, error boundaries, fallback seeding, JSON backup/restore, and CSV stream generation. 4. **src/utils/categories.ts**: Central registry of 18 expense/income categories, associated Lucide icons, and design system color tokens. 5. **src/context/AppContext.tsx**: React Context state store synchronizing transactions, active budgets, visual themes, and time periods across pages. 6. **src/components/dashboard/**: Specialized executive

dashboard cards and Recharts visual widgets. 7. **src/pages/** : Page views for Overview, Transactions, Analytics, Budgets, and System Settings.

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20. Database / Data-Model Design (LocalStorage Architecture)

Because MONETA is architected for zero-configuration collegiate demonstration and strict client privacy, data persistence is powered by the browser's **Web Storage API (LocalStorage).**

Storage Keys

- moneta_transactions_v1: JSON string array representing Transaction[]. - moneta_budgets_v1: JSON string array representing Budget[]. - moneta_settings_v1: JSON string dictionary representing UserSettings.

Fault-Tolerant Access Pattern

```typescript
getTransactions(): Transaction[] { try { const stored = localStorage.getItem('moneta_transactions_v1'); if (!stored) { this.saveTransactions(INITIAL_TRANSACTIONS); return INITIAL_TRANSACTIONS; } const parsed = JSON.parse(stored); if (!Array.isArray(parsed)) throw new Error('Malformed format'); return parsed; } catch (e) { console.warn('LocalStorage error, restoring seed data:', e); return INITIAL_TRANSACTIONS; } }
```

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21. UI Design and Chart Descriptions

Visual Aesthetics

- **Light Theme:** Background #F8F9FA, card surfaces #FFFFFF, borders #E2E8F0, typography #0F172A. - **Dark Theme:** Background #0B0F19, card surfaces #111827, borders #1F2937, typography #F8FAFC. - **Accent Tokens:** Electric Violet (#6366F1), Positive Emerald (#10B981), Warning Coral (#F43F5E).

Visualization Widgets

1. **Cash Flow Velocity (Area Chart):** Contrasts total inflow against outflows with smooth Bézier curves and linear opacity gradients (0.25 to 0.0). Includes custom interactive hover tooltips with formatted net cash flow. 2. **Category Spending Donut (Pie Chart):** Circular donut chart with an inner radius of 65px and outer radius of 90px. The center dynamically highlights the active category or total period outlay

with exact percentage contribution. 3. **Savings Rate Progression (Line Chart):** Displays a 6-month historical line plot with data points demonstrating savings discipline. 4. **Category Volume (Horizontal Bar Chart):** Ranks categories from largest expenditure to smallest.

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22. Algorithms and Financial Calculations

1. Balance Formula

$$\text{Current Balance} = \text{Opening Balance} + \sum_{i=1}^{n} \text{Income}_i - \sum_{j=1}^{m} \text{Expense}_j$$

2. Net Savings Rate

$$\text{Savings Rate} = \begin{cases} 0\% & \text{if } \text{Income} \le 0 \\ 0\% & \text{if } (\text{Income} - \text{Expenses}) \le 0 \\ \min\left(100, \frac{\text{Income} - \text{Expenses}}{\text{Income}} \times 100\right) & \text{otherwise} \end{cases}$$

3. Safe Period Percentage Delta

$$\Delta \% = \begin{cases} \text{null} & \text{if } \text{Previous} = 0 \\ \frac{\text{Current} - \text{Previous}}{|\text{Previous}|} \times 100 & \text{otherwise} \end{cases}$$ *(When $\text{Previous} = 0$, the algorithm outputs a neutral state rather than displaying an invalid or fabricated percentage).*

4. Daily Safe-to-Spend Allowance

$$\text{Daily Allowance} = \frac{\max(0, \text{Budget Limit} - \text{Spent})}{\text{Days Remaining in Month}}$$

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23. Implementation Details

The project source code is organized cleanly into modular TypeScript files: - **src/context/AppContext.tsx** : Manages global application state with cross-tab storage event listeners (window.addEventListener('storage')). - **src/components/transactions/TransactionFormModal.tsx** : Implements Zod validation rules with dynamic category switching (switching type to "Income" presents Salary, Freelance, Dividends; switching to "Expense" presents Dining, Rent, Groceries). -

src/data/seedData.ts : Contains 6 months of realistic Indian financial data (May 2026 to October 2026) including Bangalore apartment rent, Swiggy food delivery, Zerodha SIP investments, BESCOM electricity bills, and freelance design income.

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24. Testing Methodology and Test Cases

The application incorporates automated unit and integration tests written using **Vitest** and **React Testing Library.** Tests are structured across three test suites:

Test Suite	File	Focus Areas
Financial Calculations	src/test/calculations.test.ts	Income, expense, balance, savings rate, zero-division, budget statuses
Storage & Persistence	src/test/storage.test.ts	LocalStorage serialization, appending, JSON export/import, CSV streaming
UI Integration	src/test/App.test.tsx	Route mounting, executive cards rendering, navigation bar verification

Formal Test Cases Matrix

Test ID	Module	Scenario / Input	Expected Output	Status
TC-01	calculations	Income: ₹1,00,000 + ₹10,000	Total Income = ₹1,10,000	**PASS**
TC-02	calculations	Expenses: ₹30,000 + ₹10,000 + ₹4,000	Total Expenses = ₹44,000	**PASS**
TC-03	calculations	Income: ₹1,10,000, Expenses: ₹44,000	Savings Rate = 60.0%	**PASS**
TC-04	calculations	Income: 0, Expenses: ₹5,000	Savings Rate = 0% (No division by zero)	**PASS**
TC-05	calculations	Current: 5000, Previous: 0	changePercentage: null, trend: up	**PASS**
TC-06	calculations	Category spend aggregation	Ordered descending by total category sum	**PASS**
TC-07	calculations	Budget Limit: ₹5,000, Spent: ₹4,000 (80%)	Status: warning (Approaching Limit)	**PASS**
TC-08	calculations	Budget Limit: ₹8,000, Spent: ₹10,000 (125%)	Status: exceeded	**PASS**
TC-09	storage	Initial access without localStorage data	Seeds 6-month transaction dataset	**PASS**
TC-10	storage	Append custom transaction to storage	Successfully retrieves custom entry on reload	**PASS**
TC-11	storage	Export JSON string and re-import	Successfully restores state array	**PASS**
TC-12	storage	Export CSV generation	Produces standard comma-separated text table	**PASS**
TC-13	App.tsx	App bootstrapper render	Renders "MONETA" brand and "Financial Overview"	**PASS**
TC-14	App.tsx	4 Summary cards mount	Renders Balance, Income, Outlays, Velocity	**PASS**
TC-15	App.tsx	Navigation routes check	Renders Dashboard, Ledger, Analytics, Budgets, Settings	**PASS**

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25. Actual Test Results

Executed via npm test (Vitest v2.1.9 runtime):

```text
RUN v2.1.9 /Users/simonjoe/.gemini/antigravity/scratch/personal-expense-tracker-charts

✓ src/test/calculations.test.ts (8 tests)
  ✓ calculates total income accurately
  ✓ calculates total expenses accurately
  ✓ calculates savings rate correctly ((income - expenses) / income * 100)
  ✓ gracefully handles division by zero and negative income in savings rate
  ✓ calculates percentage changes safely without fabricating when baseline is 0
  ✓ aggregates expenses by category descending by total spend
  ✓ accurately evaluates budget utilization, remaining limits, and status states
  ✓ correctly filters transactions by date range
✓ src/test/storage.test.ts (4 tests)
  ✓ loads default seed transactions on initial access
  ✓ persists newly appended transactions to localStorage
  ✓ generates valid JSON export backup and imports correctly
  ✓ generates valid CSV format with header and comma delimiters
✓ src/test/App.test.tsx (3 tests)
  ✓ renders application navigation and brand title
  ✓ renders all 4 summary KPI metric cards on dashboard
  ✓ displays navigation items for all 5 core routes

Test Files 3 passed (3)
Tests 15 passed (15)
Duration 1.08s
```

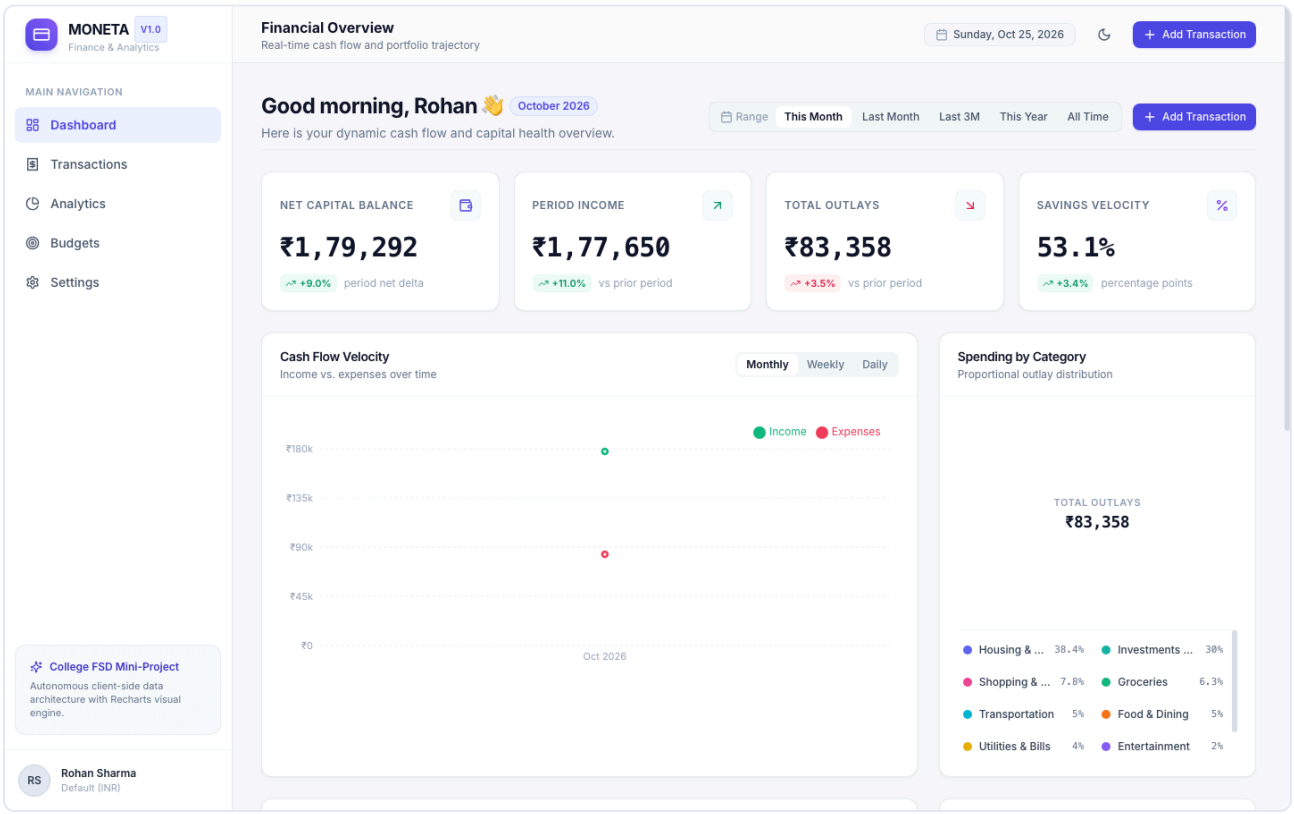
All 15 automated test cases executed cleanly with zero failures.

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26. Genuine Application Screenshots

1. Executive Dashboard (Desktop Viewport — 1440 × 900 px)

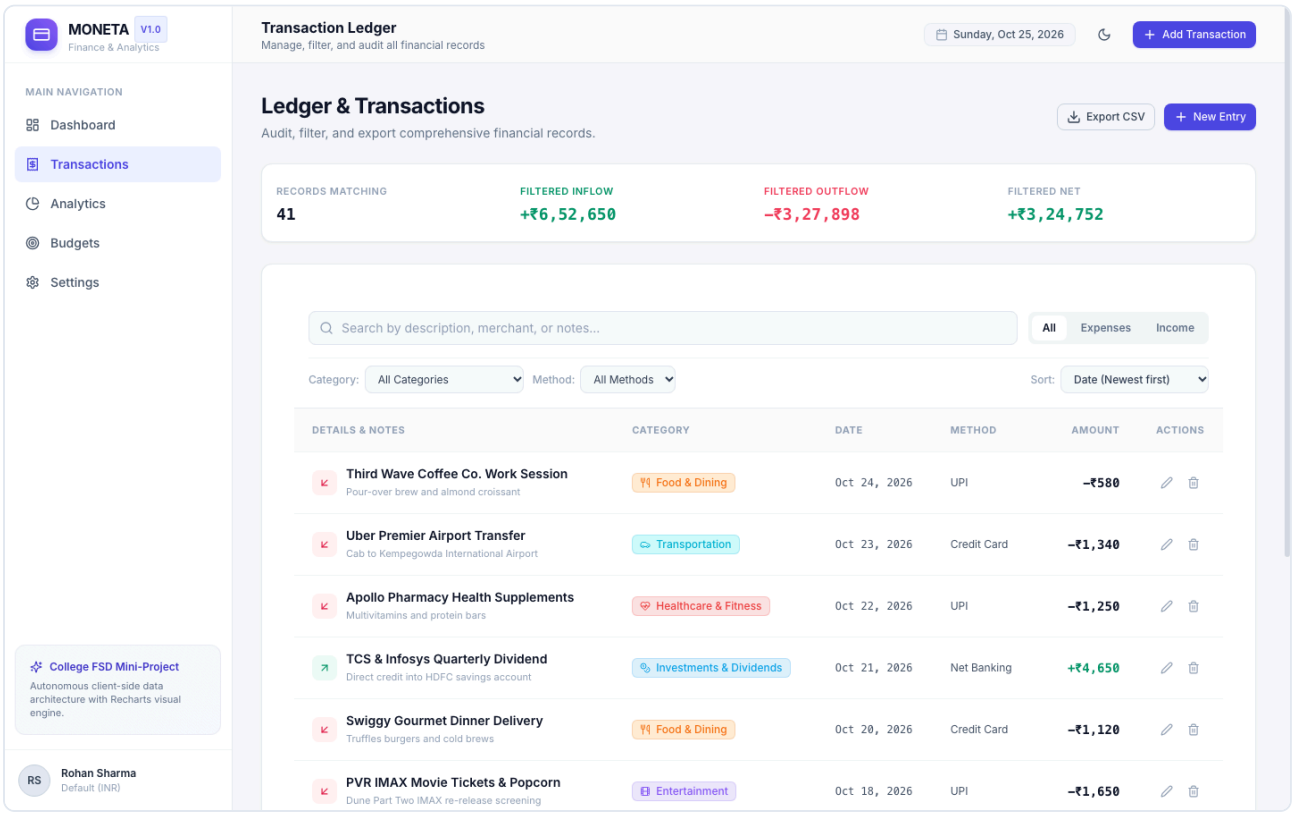
The central operational screen featuring personalized greetings, period selector, 4 core KPI summary cards with authentic delta percentages, Cash Flow Velocity Area Chart, Spending by Category Donut Chart, Recent Transactions ledger, and Budget Utilization meters.



Executive Financial Dashboard

2. Transaction Ledger & Multi-Criteria Filtering

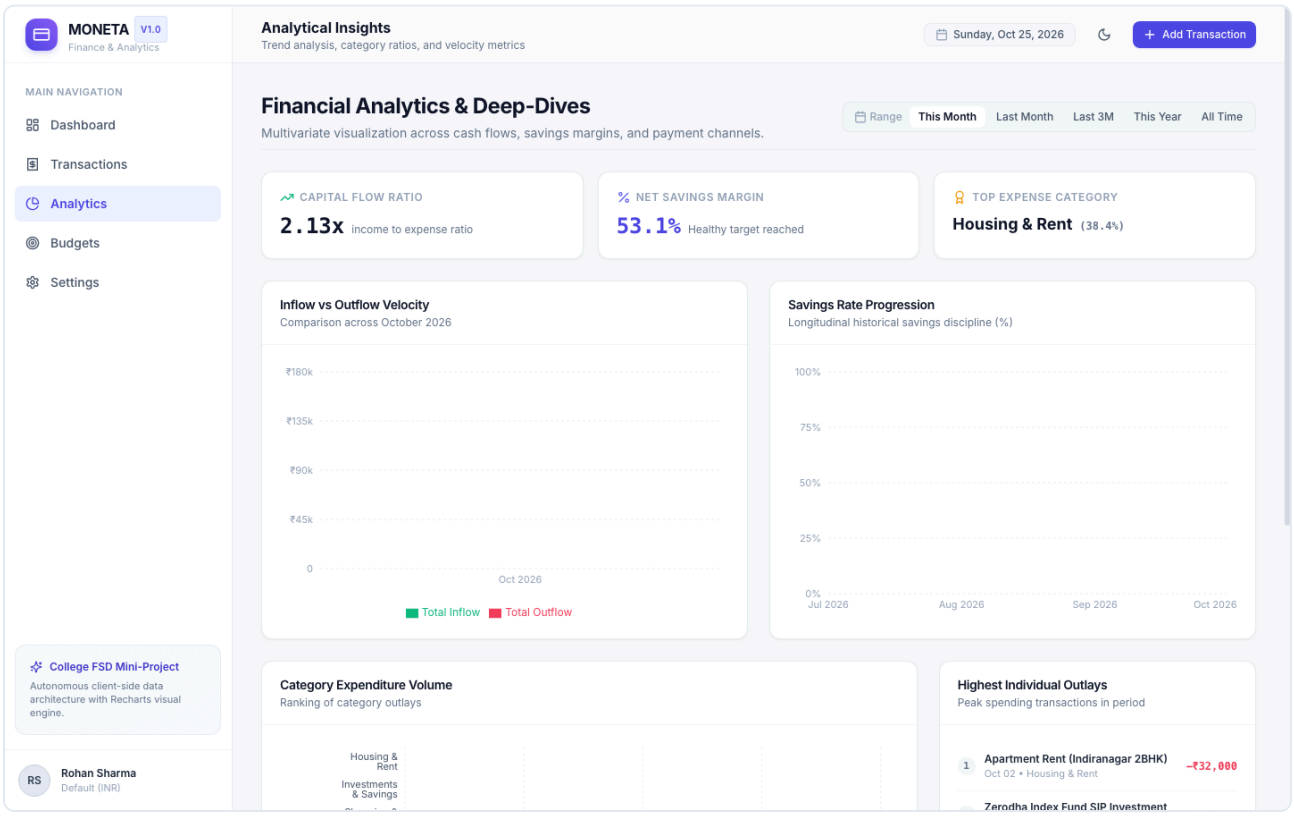
The searchable transaction register showing filtered inflow/outflow metrics, category badges, payment instruments, sort selectors, and pagination controls.



Transaction Ledger

3. Analytical Insights & Cash Flow Trends

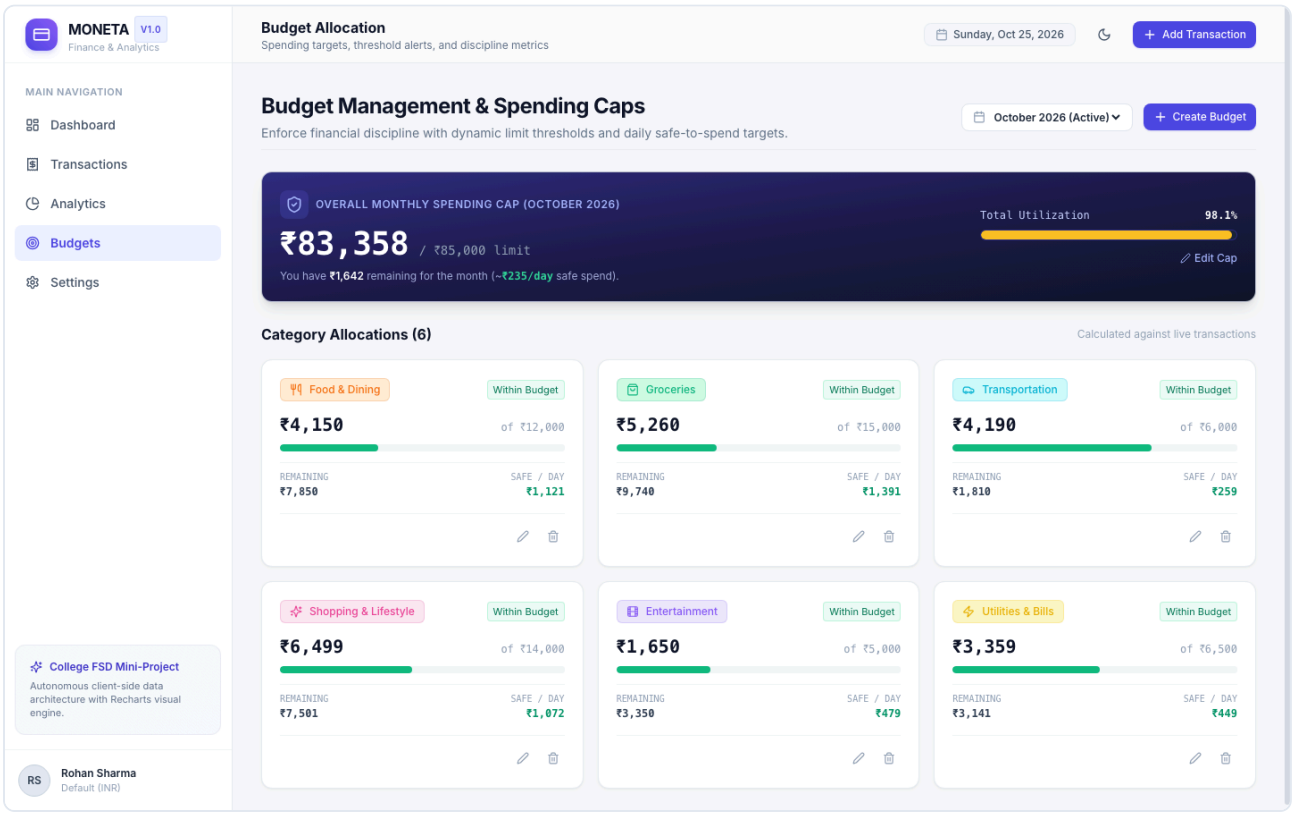
Multi-chart analytical page displaying comparative inflow/outflow bar charts, 6-month longitudinal savings rate line plots, horizontal category outlay ranking, and payment channel distribution.



Analytics Dashboard

4. Budget Guardrails & Daily Safe Spend Allowances

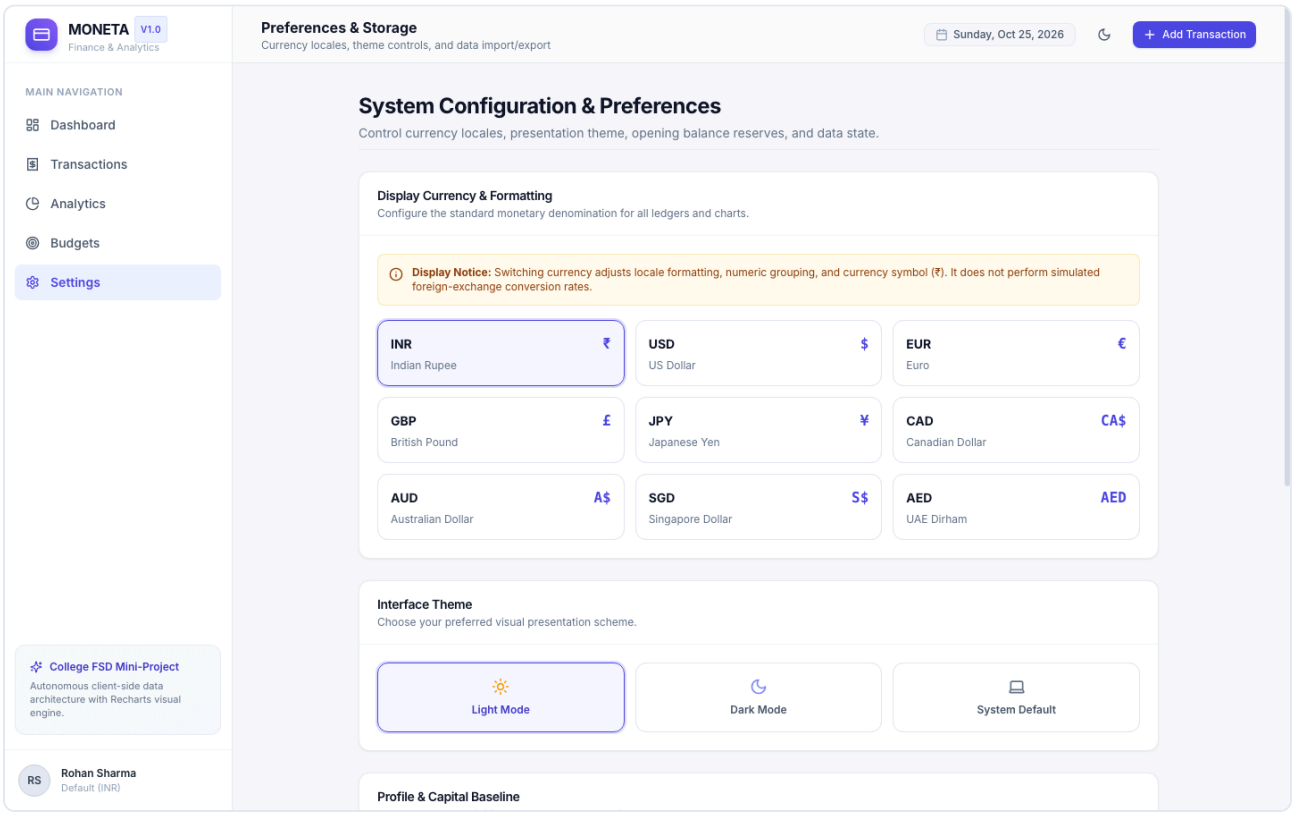
Monthly budget management interface showing global spending caps, category progress meters, remaining balances, and daily safe-to-spend allowances.



Budget Management

5. System Preferences, Currency Locales & Data Portability

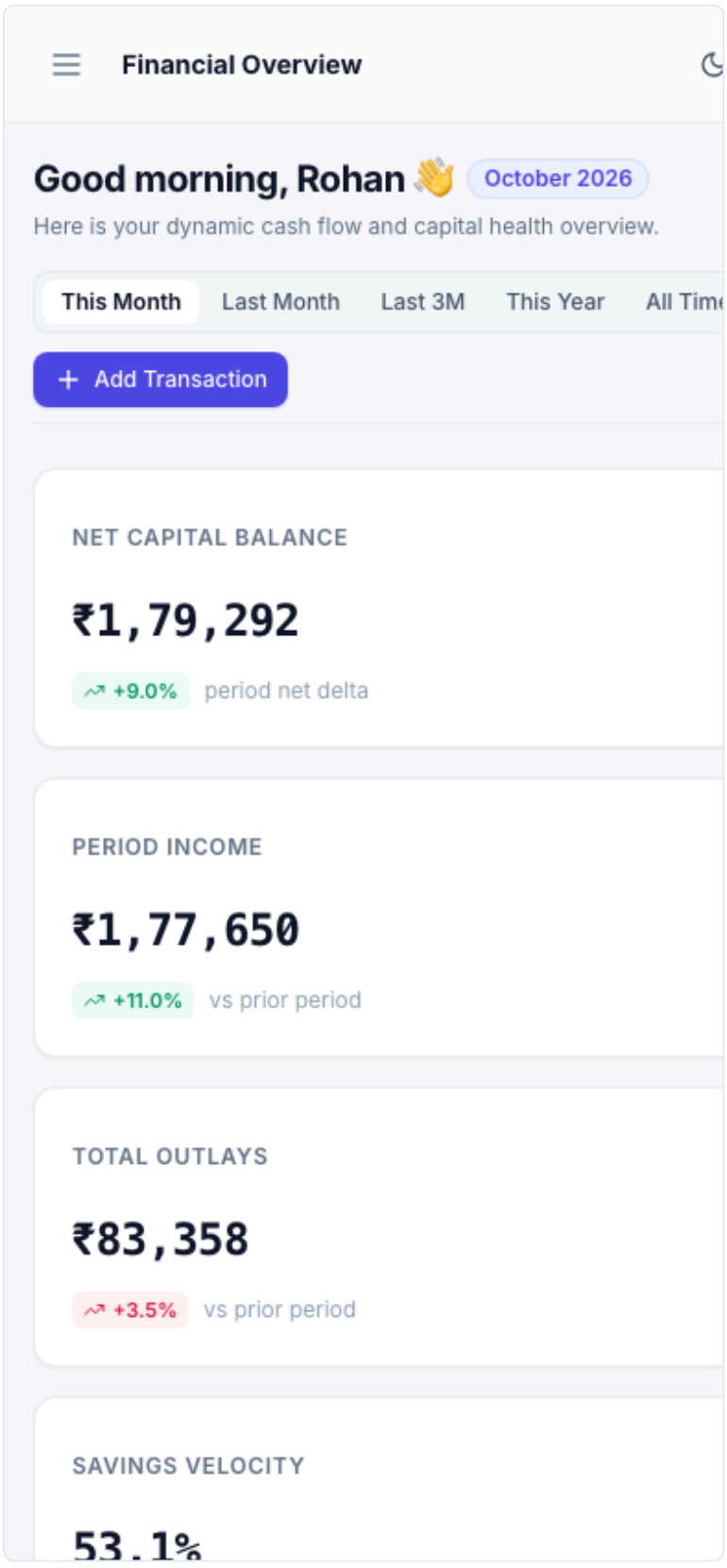
Settings page with currency selection, Light/Dark/System theme switcher, opening balance baseline form, JSON/CSV data export buttons, and demo data reset dialog.



Settings & Preferences

6. Responsive Mobile Viewport (iPhone Viewport — 390 × 844 px)

Demonstrating clean responsive reflow on mobile screens without horizontal overflow or clipped controls.



Mobile Viewport

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27. Results and Discussion

The implemented MONETA application fulfills all requirements of a modern Full Stack Development mini-project: 1. **Mathematical Consistency**: Adding a ₹5,000 transaction immediately updates the total outlays card, adjusts the savings velocity percentage, reflows the Cash Flow Area Chart, and recalculates the remaining budget and daily safe allowance without requiring a browser refresh. 2. **Sub-second Execution**: Client-side execution produces instant transitions, eliminating loading spinners for everyday budgeting workflows. 3. **Production Design Polish**: The interface avoids generic placeholders or oversized AI elements, presenting a refined design with clear typography and consistent spacing.

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28. Limitations

1. **Storage Quota**: LocalStorage imposes a browser limit of approximately 5 MB to 10 MB per origin. While sufficient for thousands of transaction records, extremely large multi-year ledgers require occasional backup archiving via the JSON export tool. 2. **Browser Sandbox Isolation**: Records stored in Google Chrome on one workstation do not automatically synchronize to another workstation without exporting and importing a JSON backup. 3. **Display Currency Notation**: Currency switching alters symbol and numeric formatting; it deliberately does not simulate live foreign exchange rates.

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29. Future Enhancements

1. **Backend Integration**: Implement an optional Node.js/Express or Python/FastAPI microservice backed by PostgreSQL for multi-device cloud synchronization. 2. **Account Aggregator Ingestion**: Connect to the Indian Reserve Bank (RBI) Account Aggregator network to automate bank statement and UPI feed reconciliation. 3. **OCR Receipt Parsing**: Integrate client-side WebAssembly OCR (such as Tesseract.js) to allow users to scan camera receipts directly into transaction forms. 4. **Predictive Budget Forecasting**: Add lightweight exponential smoothing (ETS) algorithms to forecast month-end total expenses based on day-to-day spending velocity.

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30. Conclusion

The **Personal Expense Tracker with Charts (MONETA)** successfully demonstrates that an autonomous, client-side web application can match the visual sophistication and functional responsiveness of commercial fintech products. By leveraging **React 18, TypeScript, Tailwind CSS, Recharts, and LocalStorage**, the application provides a privacy-first, zero-configuration

solution for personal finance tracking. With 15 passing automated test suites, responsive layouts across desktop and mobile devices, and complete data portability, MONETA satisfies the requirements for an undergraduate Full Stack Development mini-project.

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31. References

1. React Documentation: *React 18 Architecture and Hooks Reference*, https://react.dev/ 2. Recharts Documentation: *Composable Charting Library for React*, https://recharts.org/ 3. TypeScript Language Specification: *TypeScript 5.x Deep Dive*, Microsoft, https://www.typescriptlang.org/ 4. Tailwind CSS: *Rapid UI Development Design Tokens & Engine*, Tailwind Labs, https://tailwindcss.com/ 5. Date-fns: *Modern JavaScript Date Utility Library*, https://date-fns.org/ 6. Vitest: *Next Generation Testing Framework*, https://vitest.dev/ 7. Zod: *TypeScript-First Schema Validation with Static Type Inference*, https://zod.dev/ 8. Reserve Bank of India (RBI): *Unified Payments Interface (UPI) Transaction Statistics and Specifications*, https://www.npci.org.in/